

LAB ASSIGNMENT I

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Due: Feb 2, 2026

Instructions: Code must be written in C and well commented. Submission of code in any other file extension (.pdf, .docx etc) will not be considered. The file name of the code will be **LAST THREE DIGITS OF YOUR ENROLLMENT NO.c**

Implement the encryption as well as the decryption of the following cipher. The encryption works as follows:

1. Consider a word (without any space) as a plaintext. **(Input)**
2. Adjust the length of the plaintext and handle the repetition of letter (if present) according to the rule of the Playfair cipher. If there is any J in the word convert it to I . Let the final word be Δ .
3. Print the word Δ . **(Output)**
4. Consider a word as input and convert J (if any) to I and consider that converted word as the key K_1 . **(Input)**
5. Generate the 5×5 key matrix of the Playfair cipher from K_1 and print it. **(Output)**
6. Encrypt Δ using the Playfair cipher where the key is K_1 . Let the ciphertext be C_1 . Print the ciphertext C_1 . **(Output)**
7. Encrypt C_1 using the Affine cipher where the key is $K_2 = (11, 15) \in \mathbb{Z}_{26} \times \mathbb{Z}_{26}$. Let the ciphertext be C_2 . Print the ciphertext C_2 . **(Output)**
8. Encrypt C_2 using the Shift cipher where the key is K_3 ($0 \leq K_2 \leq 25$). Let the ciphertext be C_3 . Print the ciphertext C_3 . **(Output)**

Now you have to think on the decryption to write the code for the decryption. If the code is correct then the decrypted text after doing the Playfair cipher decryption should match with Δ . You have to print all the middle layered decrypted texts in the code.